

SIDES.GG

WHITEPAPER

v1.0 · June 2026

Live NBA Second-Screen Experience + AI Intelligence Layer

A Real-Time Sports Gaming Protocol with On-Chain Settlement and AI-Powered Engagement

Product

Live NBA Gaming Protocol

Status

Off-chain Prototype → Phase C Active

Network

Solana (SOL)

AI Layer

LLM + Real-Time Scoring Intelligence

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1. Executive Summary

Sides.gg is a real-time NBA second-screen gaming protocol that transforms passive sports viewing into an active, stake-driven competition. Users pick a side before tip-off, accumulate points from live ESPN play-by-play events, engage with co-viewers via real-time chat, and — in the full protocol vision — stake and redeem Solana (SOL) on-chain.

This whitepaper describes the current architecture, the planned on-chain staking layer, and a new AI Intelligence Layer that elevates every dimension of the product: personalized coaching, automated market-making assistance, fraud detection, and conversational game commentary.

Key Numbers

1 SOL = 200 pts · Min stake 0.005 SOL · ESPN poll interval 30s · Points scored per-action after join time

2. Vision & Product Positioning

Traditional sports betting is static: a pre-game slip, no live participation, no community. Sides.gg reimagines this as a dynamic, points-based live game layered on top of official NBA data, where outcome depends on the ongoing performance of the side you chose—not just the final score.

2.1 Core Value Propositions

- Real-time engagement: points flow on every basket, rebound, steal, and turnover — the leaderboard moves like the game clock.
- Crypto-native economics: SOL is the native unit; no fiat on-ramp inside the protocol.
- Protocol + interface separation: the Sides.gg Protocol is permissionless; this app is one of many possible portals.
- AI-augmented experience: from personalized commentary to smart staking suggestions, AI turns a game into a conversation.

2.2 Target Users

NBA Fans with Crypto Wallets

Engaged viewers who want more than passive watching; comfortable with SOL wallets and DeFi concepts.

Social Gamers

Players motivated by leaderboards, chat, and community — SOL staking is optional upside.

DeFi-Native Bettors

Users familiar with on-chain settlement who want provably fair game mechanics.

AI-Curious Enthusiasts

Users who want intelligent, real-time sports analysis integrated into their second-screen experience.

3. Current Architecture

3.1 Technology Stack

- Frontend: Next.js 14 (port 3113), RainbowKit EVM wallet connect, Supabase X/Twitter OAuth.
- Backend: Node.js / Express, Prisma ORM, PostgreSQL, Socket.IO for real-time chat.
- Data pipeline: ESPN NBA API polled every 30 seconds; normalized into Team, Game, GameAction, ActionType records.
- Identity: Supabase Auth handles X OAuth; client JWT (jwtToken) in localStorage encodes userId.

3.2 Data Model (Prisma)

Core Entities

User · Game · UserGame · GameAction · ActionType · UserStats · CodeStatus · Chat

3.3 Scoring Engine

Scoring is server-side and applied to all play-by-play actions that occur after the user's UserGame.created_at timestamp. Action weights are configured in espnConfig:

- 3-pointer made: +15 pts for the scoring side, -15 pts for the opposing side.
- 2-pointer made: +10 / -10.
- Free throw made: +5 / -5.
- Rebound, Assist, Steal, Block: +5 / -5 each.
- Turnover, Personal foul: penalty applied to committing side.
- Game winner: +30 / -30.

Conversion: 1 SOL = 200 points. P&L display at game end: SOL = points / 200 (off-chain display only until Phase C).

3.4 User Journey & Status Machine

Status	Meaning	Route
0	Registered, no valid invite	/entry — 6-digit code required
≥ 1	Invite code redeemed	/connect — Flash Staking UI
≥ 2	Side chosen, stake initialized	/chat — Live game room active

4. Roadmap

4.1 Phase Overview

Development is organized across five phases, with AI features woven into Phases B–E.

Phase	Title	Key Deliverables	AI Touchpoint
A ✓	Closed Beta	ESPN ingestion, invite-gated onboarding, off-chain points loop, live chat	—
B	Identity & Access	X OAuth hardening, wallet-user link, server-side JWT validation	<i>AI identity verification assistance</i>
C	Flash Staking	Real SOL deposits, side-pool ratio mechanics, on-chain settlement	<i>AI stake optimizer & risk engine</i>
D	Protocol & Trust	Smart contracts, third-party interfaces, geo enforcement	<i>AI compliance classifier</i>
E	Game Integrity & Ops	OT scoring, admin tools, rate limits, moderation	<i>AI moderation + anomaly detection</i>

5. AI Intelligence Layer

The AI Intelligence Layer is the primary new capability described in this whitepaper. It transforms Sides.gg from a data-display product into an adaptive, conversational gaming companion.

AI features are grouped into five modules, each targeting a distinct experience dimension: commentary, staking intelligence, personalization, moderation, and anti-fraud.

5.1 Live AI Commentary Engine

A large language model (LLM) subscribes to the same ESPN play-by-play stream already ingested by the backend. After each significant GameAction, the engine generates a one-to-three sentence commentary snippet surfaced in the /chat room as a system message.

How it works

- ESPN play-by-play action ingested (every 30 s poll or webhook upgrade).
- Action + current game state fed as structured context to LLM via Anthropic API.
- LLM returns short commentary in <200 tokens: dramatic for big plays, analytical for momentum shifts.
- Commentary injected into Socket.IO broadcast as a special system_message event type.
- Frontend renders AI messages with a distinct badge to differentiate from user chat.

Context window structure

```
{ game_state: { home, away, score, period, clock }, last_action: { type, team, player, points_delta }, user_side: 'home' | 'away', user_points_delta }
```

5.2 AI Stake Optimizer

Before a user finalizes their SOL stake on /connect, the AI Stake Optimizer analyzes the game context and provides a risk-aware suggestion. This operates fully off-chain and is advisory only.

Input signals

- Historical side-ratio data for this matchup (once Phase C pools are live).
- Current ESPN pre-game win probability.
- User's historical P&L across past Sides.gg games.
- Time remaining until tip-off (stake urgency signal).

Output

"Based on today's line and your last 5 games, a 0.25 SOL stake on the Heat gives you a positive expected-value entry. Current side ratio is 1.4:1 in favor of the Bulls — contrarian upside is meaningful."

A prominent disclaimer is displayed: AI suggestion is informational only and does not constitute financial advice. Geo-restricted users cannot access the staking flow regardless of AI output.

5.3 Personalized Game Coach

As users accumulate game history, an LLM-backed coach surfaces real-time micro-insights during /chat:

- Point swing alerts: "You just gained 25 pts in the last 2 minutes — Bam Adebayo has 3 blocks."
- Momentum warnings: "The opposing side has scored on 6 straight possessions. Consider your risk tolerance."
- Historical context: "You tend to perform best when picking the underdog in the 4th quarter of close games."

Data architecture

UserStats rows (per-action scoring) are aggregated server-side into a UserProfile vector. At each significant event the coach prompt includes: user's current game position, aggregated historical percentiles, and the last N game actions. The LLM generates personalized text in <150 tokens.

5.4 AI Chat Moderation

Real-time Socket.IO chat introduces moderation risk at scale. The AI moderation module processes each sendMessage event before broadcast:

Check	Action	Latency Target
Toxicity / hate speech	Block + warn user	< 50 ms (rule-based first pass)
Spam / repetition	Rate-limit user session	< 20 ms (Bloom filter)
Wash-trade coordination	Flag for review, escalate	< 200 ms (LLM classify)
Pump misinformation	Soft-label message as unverified	< 200 ms (LLM classify)

5.5 Anti-Fraud & Anomaly Detection

On-chain staking introduces economic attack surfaces. The AI Anti-Fraud module runs continuously against UserGame and UserStats streams:

Detected patterns

- Sybil staking: multiple accounts created via the same wallet or IP staking the same side to inflate side-pool ratios.
- Invite code farming: anomalous CodeStatus redemption patterns from automation.
- Score manipulation: unusual clustering of UserStats rows with identical action timestamps suggesting replay attacks.

- Wash-staking: coordinated stake-and-withdraw patterns between wallets on opposite sides.

Response pipeline

- Rule-based pre-filter (< 5 ms): block known bad patterns.
- LLM behavioral classifier (async): score session against fraud probability.
- Admin dashboard flag: high-risk sessions surfaced for human review.
- On-chain guard (Phase D): settlement contract checks fraud flag before releasing SOL.

6. AI Technical Architecture

6.1 LLM Provider & Model Selection

All generative AI calls route through a backend AIService module. The service is provider-agnostic at the interface layer; initial production target is Anthropic Claude (claude-sonnet) for commentary and coaching, with a lightweight rule-based pass for latency-critical moderation.

Module	Model	Max Tokens	Latency SLA
Live Commentary	<i>claude-sonnet</i>	200	< 2 s
Stake Optimizer	<i>claude-sonnet</i>	400	< 3 s (pre-game)
Game Coach	<i>claude-sonnet</i>	150	< 2 s
Chat Moderation	<i>Rule-based + claude-haiku</i>	80	< 200 ms
Anti-Fraud Classifier	<i>claude-haiku</i>	100	< 500 ms (async)

6.2 Backend Integration Points

AI services integrate into the existing Express/Prisma stack at well-defined hooks:

- `/api/game` (GET `/event`): after ESPN normalization, spawn async commentary generation for significant actions.
- `Socket.IO sendMessage` handler: synchronous moderation check before broadcast.
- `/api/user/play` (POST): trigger Stake Optimizer if user has historical data.
- `/api/user/gamescore` (POST): append Coach insight to polling response when point delta threshold crossed.
- New `/api/ai/fraud-report` (POST): admin endpoint to query fraud scores for a `userId` or `gameId`.

6.3 Data Privacy & Prompt Design

AI prompts never include: wallet private keys, raw Supabase JWT tokens, full email addresses, or third-party OAuth tokens. User identity passed to the LLM is limited to: `userId` (integer), `username`, `selected side`, and anonymized statistics.

Prompt Injection Defense

Chat messages passed to the moderation LLM are sandboxed: they are presented as data in a structured field, never concatenated into the system prompt. Output is classified via a constrained JSON schema — the LLM cannot emit free text that modifies application state.

6.4 New Environment Variables (AI Layer)

Variable	Description
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ANTHROPIC_API_KEY	Anthropic API key for all LLM calls (backend only, never exposed to client)
AI_COMMENTARY_ENABLED	Feature flag: true/false — enables live commentary injection
AI_COACH_ENABLED	Feature flag: true/false — enables in-game coaching messages
AI_MODERATION_ENABLED	Feature flag: true/false — enables chat moderation pipeline
AI_FRAUD_ENABLED	Feature flag: true/false — enables fraud detection async worker
AI_COMMENTARY_THRESHOLD	Minimum point delta to trigger commentary generation (default: 10)
AI_MAX_COMMENTARY_PER_MIN	Rate cap for AI messages in chat per game (default: 4/min)

7. On-Chain Staking — Phase C

Phase C delivers the core economic promise of the product: real SOL moves on-chain before tip-off and settles automatically after the final buzzer.

7.1 Flash Staking Mechanics

- User stakes SOL into a per-game side pool (home or away) before tip-off.
- Points accumulate per the existing scoring engine during the game.
- At game end, the winning side's pool is distributed proportionally by final points — not equally by stake.
- Side ratio = (winning pool total SOL) / (losing pool total SOL); published on-chain pre-game.

Example

Home pool: 100 SOL · Away pool: 60 SOL · Ratio: 1.67 : 1. User stakes 1 SOL on Away, earns 8,000 pts, total away pts = 100,000. User's share = 8% of 60 SOL = 4.8 SOL returned. Effective return: 4.8x.

7.2 Settlement Contract

- Solana Program (Rust / Anchor): trustless custody of both side pools during the game.
- Oracle feed: backend game-server posts signed final score + winning side after ESPN confirms FINAL status.
- Users call redeem() after settlement; contract verifies signature, calculates share, releases SOL.
- AI Anti-Fraud flag: contract checks a fraud_score threshold; flagged accounts enter a 24-hour hold for human review.

8. Implementation Status & Gaps

The table below shows the full feature surface against current build status and planned completion phase.

Feature	Current Status	Target Phase
ESPN ingestion & normalization	✔ Implemented	Phase A ✓
Play-by-play scoring engine	✔ Implemented	Phase A ✓
Invite code gating	✔ Implemented	Phase A ✓
Off-chain points loop	✔ Implemented	Phase A ✓
Real-time chat (Socket.IO)	✔ Implemented	Phase A ✓
X (Twitter) OAuth	❑ Requires Supabase config	Phase B
Wallet ↔ User record link	❑ Partial (UI only)	Phase B
Server-side JWT validation	✗ Client JWT only	Phase B
Real SOL staking	✗ Demo mode	Phase C
Side-pool ratio mechanics	✗ Not built	Phase C
On-chain settlement	✗ Not built	Phase C
Smart contracts (Anchor)	✗ Not built	Phase D
Geo/sanctions enforcement	✗ Disclaimer text only	Phase D
AI Live Commentary	✗ Proposed	Phase B/C
AI Stake Optimizer	✗ Proposed	Phase C
AI Game Coach	✗ Proposed	Phase B/C
AI Chat Moderation	✗ Proposed	Phase E
AI Anti-Fraud	✗ Proposed	Phase D/E
OT / correction scoring	✗ Not built	Phase E
Admin tools & monitoring	✗ Not built	Phase E

9. Legal & Compliance

The application includes a mandatory /disclaimer route. Users in prohibited jurisdictions — including the United States — cannot access staking functionality. The Sides.gg Foundation operates the protocol; the foundation does not custody user funds once on-chain settlement is live.

9.1 AI-Specific Compliance Obligations

- AI-generated commentary and coaching output must be labeled as AI-generated in the UI.
- AI Stake Optimizer output must carry a prominent disclaimer that it does not constitute financial or investment advice.
- Anti-fraud model decisions that restrict user access must be auditable and subject to human review within 24 hours.
- No AI model may be used to infer or reveal personal information about third parties (e.g., identifying real players from chat descriptions).
- All AI vendor data processing agreements must comply with applicable privacy regulation in operational jurisdictions.

Operators must independently verify that product, AI model deployment, and marketing practices comply with all applicable laws and regulations in each jurisdiction they serve. This whitepaper does not constitute legal advice.

10. Conclusion

Sides.gg has a solid off-chain prototype: ESPN data ingestion works, the scoring engine is live, and the onboarding flow is functional. The path to a full protocol is clear across five phases.

The AI Intelligence Layer is the differentiating leap. It converts raw play-by-play data into narrative, converts stake decisions from guesswork into informed choices, and converts an open chat room into a safe and engaging community. Together, these features transform Sides.gg from a clever concept into a defensible, high-engagement product.

Next steps

1. Complete Phase B: harden X OAuth and link wallet addresses to backend user records.
2. Integrate AI Commentary & Coach behind feature flags — ship to closed beta for feedback.
3. Begin Anchor smart contract development for Flash Staking (Phase C).

Appendix A: API Overview

Base URL: {SIDEKG_SERVER} — typically <http://localhost:4444/api>

Game Routes (/api/game)

Method	Path	Purpose
GET	/event	Refresh/process ESPN data
GET	/raw	Raw ESPN payload (debug)
POST	/search	Game by gameId
POST	/actions	Play-by-play for game
POST	/score	Team score in game
POST	/clock-period	Period + clock
POST	/initialGame	Game list/detail for UI

New AI Routes (/api/ai)

Method	Path	Purpose
POST	/commentary/generate	Force-trigger commentary for a gameId (admin)
POST	/stake/suggest	Get AI stake suggestion for userId + gameId
GET	/coach/insight	Poll latest coach message for userId + gameId
POST	/fraud/report	Query fraud score for userId or gameId (admin)
GET	/moderation/status	Health check for moderation pipeline